

CST8504 Lab5 Create3 ROS2 - Submission Answers

Python Playground Submission

Submission 1.1: What does the comment say about the line of code

```
robot = Create3(Bluetooth())?
```

Answer:

The comment says: "**Will connect to the first robot found.**"

(Found in create3_all_options.py line 10: `robot = Create3(Bluetooth()) # will connect to the first robot found.`)

Submission 1.2: What color are the lights set to when the right bumper is pressed? (read the code to find out – try it if you're stuck)

Answer:

The lights are set to **green** when the right bumper is pressed.

RGB values: `(0, 255, 0)` - green

(Found in create3_all_options.py lines 31-34:

```
@event(robot.when_bumped, [False, True])
async def bumped(robot):
    print('Right bumper pressed')
    await robot.set_lights_on_rgb(0, 255, 0) # green.
```

)

Submission 1.3: When you run the create3_all_options.py file, what information about the Create3 is printed starting with http (http, Name, Battery, Serial #, etc)?

Answer:

When you run create3_all_options.py, it prints the following information about the Create3:

1. **http://[IP address]** - The **robot's** IPv4 address (e.g., `http://192.168.186.2`).
2. **Name:** - The robot's name
3. **Version:** - The robot's version string

4. **Battery:** - Battery level in mV and percentage (e.g., `Battery: 15000 mV; 100 %`)

5. **Serial #:** - The robot's serial number

6. **SKU:** - The robot's SKU (Stock Keeping Unit)

(Found in `create3_all_options.py` lines 90-101:

```
address = await robot.get_ipv4_address() # Gets the robot's IP address
ip = address.wlan0
print('http://' + str(ip[0]) + '.' + str(ip[1]) + '.' + str(ip[2]) + '.' +
      str(ip[3]))
battery = await robot.get_battery_level()
print('Name:', await robot.get_name())
print('Version:', await robot.get_version_string())
print('Battery:', battery[0], 'mV; ', battery[1], '%')
print('Serial #:', await robot.get_serial_number())
print('SKU:', await robot.get_sku())
```

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